Aditya Gandhi

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Education

Purdue University, West Lafayette **August 2023 – May 2027**
Bachelor of Science (B.S.) in **Computer Science Honors**
Concentrations: Software Engineering, Machine Intelligence; Minor in Mathematics
Coursework: Analysis of Algorithms, Competitive Programming, Systems Programming, Data Mining & ML, Robotics
Activities: Teaching Assistant (Analysis of Algorithms), Developer for Hello World (Purdue's Largest Hackathon)
Achievements: AMD Track at StarkHacks, Alpha Lambda Delta Honor Society, Dean's List

Experience

Amazon June 2026 – August 2026
Software Development Engineer Intern Bellevue, WA
- Designed and productionized **105+ native Kotlin UI tests** covering the two largest screens on Amazon Flex **(400K+ users)**, replacing manual suites in a daily release pipeline used by **40+ engineers**.
- Built reusable Kotlin UI-test infrastructure with workspace-aware navigation and actionable failure diagnostics, reducing repetitive setup and manual intervention.
- Standardized test execution and resolved framework limitations, cutting a 40-test validation cycle from **8 hours to at most 20 minutes of automated testing**.

Duality Lab January 2026 – May 2026
Undergraduate Researcher (Systems) Purdue University
- Led a 5-person team to build a **resource-constrained runtime** for OpenSWE, enforcing token, tool-call, and wall-clock budgets during execution.
- Implemented configurable routing and graceful degradation policies, recovery across **12+ failure categories**, and per-node telemetry for observability.

Retailmind June 2025 – August 2025
Software Engineering Intern (Distributed Systems)
- Architected a distributed, multithreaded backend processing data across **4,500+ store–department entities**.
- Designed asynchronous pipelines that maintained availability under compute-intensive peak loads.
- Engineered a hierarchical forecasting service with **sub-300ms inference latency** across multiple model backends.

Indegene June 2024 – August 2024
Software Engineering Intern KA, India
- Built Python services for multi-hundred-page regulatory-document analysis and source-backed article generation from structured Excel data, streamlining compliance and content workflows.

Projects

HypeTrade | Python, FastAPI, GCP Cloud Run, Redis, Docker, Cloud Build, React
- Architected and deployed FastAPI services on **GCP Cloud Run** supporting **500+ concurrent users** with **99% uptime** and sub-300ms API response times.
- Led 6 developers in delivering authentication, portfolios, messaging, and a real-time discussion forum.
- Automated container deployments through Cloud Build CI/CD, reducing build time by **90%**.
- Optimized batched FinBERT inference for 500+ daily tickers, reducing model-serving costs by **56%**.

Systems Programming Projects | C, C++, Flex/Bison, x86-64 Assembly, Linux
- Implemented a single-pass **SimpleC-to-x86-64 compiler** with Lex/Yacc, generating code for expressions, arrays, functions, and control flow.
- Built a **Unix shell** with Flex/Bison parsing, pipelines, I/O redirection, background processes, built-ins, subshells, expansions, signal handling, and command history.
- Implemented x86-64 heap sort with recursive heapify, frame-pointer stack unwinding with signal handlers, and hexadecimal memory/file dump utilities.

Technical Skills

Languages: C++, Java, Python, C, Kotlin, SQL, TypeScript, JavaScript, Bash
Systems & Backend: Linux, FastAPI, Node.js, Redis, REST APIs, Gradle, Git, Flex/Bison, x86-64
Cloud & Tools: Docker, GCP, Cloud Build, CI/CD, Kubernetes, React, PyTorch, Hugging Face, Pandas, NumPy